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H-SCOOP

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by Henry Fale

BITS & PIECES

WHAT'S NEW WITH H-SCOOP?

As you may have noticed by now, the name of this newsletter has been changed from H8SCOOP to H-SCOOP. The old name was too confusing as it did not reflect my support of the H89. I'm sure many subscriptions have been lost because of that. This will also allow me to handle any of the other Heath H- line in the future and have the newsletter name reflect that.

You may have noticed in the December issue that the Subscription rate has been increased to \$20. This was necessary because of an increase of all my operating costs, especially postage which went from 15 to 20 cents since my last price increase. Although the \$20 rate was posted in December, these rates will not take effect until February. Those who have already sent in \$20 have received the appropriate amount of extra issues.

This year a few things will change. Since **NEW STUFF** is getting so large, I will be limiting the information in that column so the entire newsletter doesn't end up being **NEW STUFF**. I will more carefully edit all materials sent to me, and print what I feel is important for the reader to get the general idea of what it is and does, and then leave it up to the vendor to supply additional information if requested. This way I can still give a good mention, though not in extreme detail to take up lot of space. I don't like to have to edit things down that much, but I feel it has come to the point where it is necessary to preserve the general format of this newsletter.

I also plan to do more reviews this year, especially software, since many individuals are concerned about getting burned. This may help prevent piracy to some extent. I will mainly review what is submitted to me. If you have something you feel good about, send me a copy for evaluation. Keep in mind, I will be honest about the findings. I will print the good as well as the bad. Since I don't have the time to review everything, I have worked out with a fellow computerist to help on the reviews. This means anything submitted to me MAY be copied one time only, to allow me to send him a copy for review. You usually can't go wrong with this type of deal because the only cost you're out is the cost of a disk. Be assured that no additional copies will be made or distributed. And since this takes time, don't expect to see a review right away.

One of the problems with a micro computer is limited storage and memory. This has reflected on my mailing list program, which has taken many disks to hold all the names. I now have a mailing list program using an 8" drive where I can have 30,000 names on line at one time, and will be changing over to it early in 1982. I have to iron a few bugs out of it yet. Once it is done and on line, the mailing portion will be very modern and fast. I may make this program available; it runs under CP/M. I'll keep you up to date on it.

I will also be extending the **GOOD GUYS & BAD GUYS** column to include individual Heath stores. I receive much information and complaints about Heath stores. I have been receiving lot of complaints of late about poor service and broken promises by Heath stores, which I feel is inexcusable. If Heath wants to sell their systems as business systems, they had well better be able to provide the necessary support, or close their doors! In the business world, if a system is down for a day, that's a long time. I have reports of business systems being down for weeks because of the inability of some Heath stores, and this will be printed. The reason for this is twofold. First, to let you

know who the good stores are, the ones that will support you, and those to stay away from. Second, if they (and Benton Harbor) see their names and reports in print, they may take some care to straighten out the problems to avoid further embarrassment or lost service. So feel free to send in comments and experiences about your local Heath's and HUG's. Instead of Good and Bad guys, these stores will receive the **DISK STRUCTURE CORRUPT**, or **DATA SEEMS INTACT** mentions. If I happen to think of better names by then, these may be changed.

As usual, any comments on this newsletter, content or format, will be taken into consideration. If you have something you like or don't like about it, write me. I depend heavily on reader feedback to keep **H-SCOOP** the way you want it.

CREDIT OMISSION

In October's issue under Technical Forum, I printed an article on "Customized Bootable Disks" and gave credit to CHUG, which is where the article came from. I neglected to mention the author. This was an excellent article, and I want to give credit where it is due. This was written by John Stetson. My apology for this omission.

BYE BYE MARK GORDON COMPUTERS

Bob McClure Jr. sent me some information about Mark Gordon Computers, the ones I mentioned several times in the **BAD GUYS** column. It turns out they have filed for bankruptcy. If you have been burned by them, you must file a claim before June 7, 1982 in the office of the Clerk of the Bankruptcy Court/ 212 John W. McCormack Post Office and Courthouse Bldg/ Boston, MA 02109. If you need further info, contact the U. S. Trustee, Richard Kagan Esq./ 89 State St/ Boston, MA 02109.

PRAISE FOR H-19

Ralph Munroe sent me some clippings from Personal Computing, November 1981. On page 7 is an article praising the H-19 for its low cost and "standard". I quote, "If there is a 'standard', it is the H-19 from Heathkit. . . The H-19 is a smart terminal that cannot be duplicated for less than \$1500, yet the H-19 costs less than virtually all dumb terminals on the market. . . It may cost like a Toyota (yea--my car--ed), but it is designed like a Mercedes. . . According to the Boston Computer Society, MIT students are buying H89s and H8s. Why? Heath equipment is designed like a Mercedes. The operating system, HDOS, makes CP/M look like a dinosaur (AMEN to that--ed). The documentation is unquestionably the best in the business. . .". Also in that issue, page 20 is some info on new tax laws that affect computer owners using their systems for extra income--something you should read if you are in that boat.

MISC FROM RICHARD HARTMAN

Richard writes in about the "Y" connector for the disk cable mentioned in December's issue. "If the connector on the disk controller is replaced with a right angle connector, only one slot is needed for hooking up two cables. The solder pads are small and care must be taken in desoldering. Fortunately, most of the circuit is on the back of the board making repair easy." Very good idea, I never thought of that.

As for the **MPI DRIVES**, Richard says he's been using an MPI-52 for over 6 months without any problems. Many individuals feel I am against the MPI because I am handling the Tandon. NOT TRUE! I am handling the Tandons because of what I have heard about the MPI. I may have

been too harsh with MPI, but when I was investigating this whole matter many months back, I called several distributors about my handling different drives. Several told me that they were discontinuing them and going to Tandons because of problems and bad service. One place even offered to sell me his entire stock at cost just to get rid of them! I also understand in evaluations, Heath rejected the MPI for the Tandons, which they are currently handling. You may have noticed, IBM also choose the Tandons for their new computer. I feel that speaks for itself and needs no further clarification or defense on my part. I will add, however, that indications seem to be that MPI has cleaned up their act somewhat and things may be better now than they were 6 months ago.

Richard also has the highest praise for MAILPRO, a random I/O mail program from Clark Systems (see past H8SCOOPS).

RE-ENTER H47?

Informants tell me that the H47 is to be re-instated and back in full force again. What happened? I don't know. Between the too high price, and the unreliability problem, it was doomed from the start. If the price was lower, it may stand a chance, but I'll still take my Magnolia system over the H47 any day.

SUPERCAL BITS

Yes, SuperCalc is here, and it's a great package. I've been playing around with it for some time, and am very impressed. What I stated last month is true--it cannot be PIPed to a different format disk, only DUPed to the same format. Is that the end? NO! Magnolia Microsystems, bless their sweet hearts, has a conversion program for \$25 that will allow copying SuperCalc from any format to any format. I know. I purchased it in the 5" SS SD and converted it to my 8" DS DD, no problems. If the SuperCalc is purchased from them at \$295, it is thrown in with the package if requested.

HDOS BB SYSTEMS UP

I have a list prepared by Tom Jorgenson on the present listings of HDOS bulletin board systems (HBBS). Since it is too much to print here, if you want a copy, send a SASE plus \$1 for copy and handling. This list will be updated from time to time as new changes become available.

HDOS MBASIC COMPILER PROBLEMS?

I have received information that some bugs exist in the HDOS MBASIC Compiler being distributed. Microsoft requires a \$40 royalty on every computer system that is running the new system. This means if you write business code and compile it into the .ABS file for someone else to run, they must legally send Microsoft \$40. It is also reported that of the known bugs, Microsoft has no intention of fixing them for current purchasers. To get the fixes you'll have to buy the new version when it's released. Since Heath does not know the internals, they are really not in a situation to help either. You may want to think twice before purchasing this release, or the compiler in general! This information was taken from SPOHUG newsletter, Editor Charles Ballinger/ RFD 1 Box 676/ Spokane, WA 99204 (509) 448-9727.

RS-232 DISTANCE

What is the allowable distance for devices connected to a computer via RS-232 cable? This has been the subject of many requests in past issues of this newsletter. BYTE magazine, December 1981 page 253 had some answers. It seems that if operating up to 9600 baud, 50 feet should be no problem. If running at lower baud rates like 1200, you should be able to separate the computer from the peripheral by as much as 500 feet. There are special cables available to get the maximum distances. These are low capacitance cables, and some even have every lead

shielded! Expensive? Yes, but they do the trick.

TANDONS WITH HEATH H8 Z80

There have been some problems with getting the Tandon 400K drives to work with the new Heath Z80 CPU. The first problem is side select. Although the board does support it, Heath does not! They refuse to give any hints on how to use the side select on this board (nice guys!). Referring to the 0-org manual, you have to connect the center of the 3 pin jumper to pin 32 of the H17 disk controller cable plug located on the H17 controller card. This is done on the foil side of the H17 card. It is important to note that the pin numbers are even on the top row, and odd on the bottom row. This is the most common wiring mistake--to mess this up.

More seriously, there has been a problem with the SY.DVD from UltiMeth not working with this card, even though it worked with the 0-Orig card. Dean Gibson of UltiMeth, with help from HUG's Pat Swayne came up with the answer on that.

It seems that when the system comes up on the 8080 card with the 0-org card, ROM is copied to RAM and you can still address from 4K and up. Heath decided to do things differently with the Z80, which only allows addressing from 8K up, and thus the problem. To make the device driver work (DKH17V2) the following patches must be made. Note this is for the version 2 release only.

Sector 8 (on 0-9 actually sector 9) address starting with 91 should read 32 36 20 (hex). Change these values to 00 00 00. This will allow you to INIT your disks. If you have already INITed your disks with other than the Z80 and now want to use the Z80 you can make the same changes to the boot track of your INITed disks as follows. Track 0 sector 1 address starting with 8B should find 32 36 20, and again, should be changed to 00 00 00. This will allow them to work with the Z80 board.

If you have version 1 SY.DVD the changes are the same, but the locations are different. For the first part, check sector 6 and address 61. For the second part (disks already INITed) the boot track location of track 0 and sector 5B should give you the proper results. This should solve your problems with SY.DVD. Anyone who received a copy of the driver after December 1st, will have this taken care of on the disk.

H8 SUPPORT

Disappointingly, HDOS 3.0 soon to be released is said to be for the H89 only, and will not work with the H8. The H8 appears to be doomed from Heath, but not to worry. I have good news that Trionyx will continue to support the H8, even be producing them when Heath stops. Of course D-G Electronics will also be supporting them, along with others like MI-8 and Environmental Systems. I have received a further commitment from Ray Livingston of Livingston Logic Labs for his continued support of the H8. Ray already has the 8" controller for the H8 and is going to expand the controllers for the H8. It is important to note that these are for H8's with Z80 CPU only. Before making any commitment, Ray wants feedback from H8ers. He will go into production on the 8" DD version only if he is assured a certain amount of sales. Write Ray if you are interested, at POB 5334/ Pasadena, CA 91107, or call at (213) 792-4763.

What is presently planned by Ray is a PC17 double density 5" controller. This will be a replacement board for the present H17 card and will allow both single and double density. At single density, it is 100% compatible with the existing board and software. Either single density or double density operation is available when used with PC17 software drivers with HDOS or CP/M. The card will support 4 drives in any combination of 40 or 80 track, single or double sided, and single or double density. The maximum storage is doubled to give 200K on a standard

40TK SS drive, and up to 800K on the double side, 80 track drives. That is more capacity than with any other Heath controller. Automatic detection of disk density is provided, allowing both single and double density disks to be used in the same drive. This will be released around February.

The boards will be sold assembled and tested with choice of HDOS or CP/M drivers included. Approximate prices are \$300 for H8 and \$275 for H89 version. (There will also be one of these cards available for the H89). PC17-8 for the H8, and PC17-89 for the H89.

Ray is also planing for the H8 a FDC-Z800 disk controller. It will be double density soft sector and will handle 4-5" (any type) and 4-8" drives. With the 8", standard IBM 3740 soft sector format will be provided for complete compatibility with the Heath H-37 and H-47, as well as the entire CP/M standard world. The estimated price of this will be about \$450, write Ray if you have an interest.

Ray also wants me to remind you that he has the single density, CP/M compatible controllers for both the H8 and H89 as mentioned in the past few H8SCOOP's. Contact Ray for full info on this. I have had reports from some who have purchased these controllers, and they are very happy with them. It seems to be a product which can be highly recommended.

HEATH/ZENITH STUFF

Many of you may have received the Heath/Zenith Software catalog for the H8, H89 and H89 computers by now. It is quite well done, containing their entire line of business and applications software, including the Softstuff line. If you have not received it, contact Benton Harbor and request Mail Order Catalog #815.

I received some info on the new Heath super computer which a subscriber got a peak at. He reports it is a 16 bit machine with dual horizontal Tandons, and a larger keyboard. It looks something like an Apple configuration in beige tones with what looks like a separate monitor on top, but it may be a single unit, he did not know for sure. The keyboard unit probably houses the main CPU electronics and the top monitor part houses the drives.

It looks like Heath also will have the CP/M educational course (EC-1120) for \$99.95. The H19 to 89 mod (H19-2) is lowered to \$495, and the H19A to 89 mod (H19-3) is also \$495. The H19A has more shielding and the video board is split in two to minimize RFI. The H89 multi mode interface card (Z89-11) will sell for \$150, with software for \$895 to emulate IBM terminals. The board will have 1 parallel (for driving MX-80) and 2 serial ports, one of the serial ports configurable for various modes to communicate with IBM terminals 3780, 2780, 3741, 2770 and remote batch terminals.

HDOS 3.0 to be released in March for \$150 available in different formats for 5" hard or soft and 8" hard or soft sectoring.

HEX OUTPUT ON H8 FPLEDS

I have a routine sent in from D. C. Shoemaker that will force hex output on the H8 front panel. When combined with the modified HDOS Ver 2 assembler and cross reference generator, you have a very fine assembly package, well suited to someone who likes hex, or who wants to type in a program listing that was created on a hex system. I will not print this here because it will take up more space than I have, and I do not know if many are interested. If you are, send me a legal size SASE and I will send you a copy.

CP/M FORMAT HINT, H37 HINT

This would normally go in technical forum, but because of lack of space this month, there will no no Technical Forum

section. Ray Livingston tells me that when you format under CP/M, by saying `FORMAT F` or `FORMAT *`, no media check will occur and formatting a disk will be very fast. Without this option, normally 4 writes and reads are done.

Ray also tells me that although not documented, the H37 can be set for extended density to get extra storage, equal to double the single density storage, or in other words, 200K for the 100K drives. To do this, you can type in an `E`, for density instead of a `S` or `D`.

NEW STUFF

H89 BOARDS

Ray Alberktson/ 619 W Dover/ San Bernardino, CA 92407 (714) 882-8072 has completed a prototype of a Sound/Clock/Calendar board for the H89 and is planning to sell the bare boards for \$22. It can be populated for about \$30 more. It uses the AY3-8910 sound generator, with the usual 3 tones and noise channels, mixer, envelope control etc. The clock is the MSM5832 buss-oriented clock chip, providing seconds, minutes, hours, days, months, and year, all easily accessible under software control. The board must be on the right side of the H89, and uses either the DTE or DCE address at 320 or 330. Included is complete assembly instructions, parts suppliers, and sample MBASIC programs for setting and reading the clock/calendar. \$5 and SASE will buy the doc kit-all you need to wire-wrap your own. To keep costs down, this board is single sided and NOT solder masked or silk screened.

For \$5 a Sound Editor program is available with purchase of the above which enables the user to experiment with all facets of the PSG in real time. The program is supplied on 5" HDOS disc, requiring 48K and supports up to 2 PSG chips, such as the Mako Data Products Sound Board. It also uses H89 graphics. Cost is \$15 without board purchase. By the time you read this, they should be ready for shipping.

From **FRONT LINE COMMUNICATIONS, INC.**, (see latter) comes a **REAL TIME CLOCK** for the H89 and H8. Time of day to a millisecond, in 12 hour and 24 hour format. Day of the week, date, month, year. On board NICAD will keep the clock running for more than 2 months. Software (source code) to set and read clock include^d. Compatible with CP/M 2.2, and HDOS 2. H8/Z80 version is \$142.50 and H/Z89 version is \$119.

Real Time Clock Software option allows you to call up to 6 programs from disk at times you preset. If the computer is working on something else the time call can be postponed or interrupt the program. Z80 version only, for CP/M or HDOS 2.0, supplied on 5 or 8" disk, for \$29.95.

The **Z89-11** will be available from Heath for the H89. What is that? It's a multi I/O card supporting synchronous and asynchronous serial I/O (2 channels), standard or IBM 3780 remote batch terminal processing (with proper software). It also has a Centronics compatible parallel 8 bit port for driving MX-80 or whatever. (See Bits & Pieces). Expect to see it the first quarter of this year.

NTS COURSE INDEX

Available from **H Moxness**/ Rd 6 Box 176/ E Stroudsburg, PA 18301 is **NTS Microcomputer Course Pamphlet Index**. For anyone taking the National Technical Schools Master Course in Microcomputers, which culminates with the building of an H89, an **ALPHABETICAL INDEX** to the 59 main course pamphlets is available for \$5.50 from Mr. Moxness. It contains 898 entries and is of great value in making the bulk of the course material usable as a reference text. It will help you when referring back to previous course material.

UCSD PASCAL HANDBOOK

From **GINGER BRED SOFTWARE**/ POB 1627/ Santa

Barbara, CA 93116 (805) 685-6556 (after 5PM CA time) comes **UCSD PASCAL HANDBOOK**. A Four part handbook of step by step procedures on how to use the Softech Heath UCSD Pascal. Part I is complete and includes introduction, overview of the system, and procedures on how to configure your working disks and use of the FILER. It is available for \$3.50 plus \$1.00 shipping. Handbook is unbound stapled form. If you desire it in a folder, include \$1.00 more. CA residents add 6% sales tax.

They also have available for \$15.00 plus shipping a **COMMUNICATIONS** package for the H89 user of version II.0. No programming necessary and detailed manual for use is included. For more info on these, contact Gene Gingerich of GINGER BRED SOFTWARE.

GENERAL FIND ALL INDEX

MISSOURI INDEXING, INC./ POB 301/ St. Ann, MO 63074 (314) 997-6470 has available a new article finding guide called **THE INDEX**.

THE INDEX is a unique information resource compiling over 12,000 articles, editorials and columns from over 900 issues of personal computing magazines written in the last 6 years, into over 30,000 entries.

Its purpose is to index by subject the massive amount of information that has always been available, but practically impossible to gather and maintain. A method of indexing called "Key Word in Context (KWIC)" was utilized. This indexes an article not only by subject, but by key words for a specific subject. This index not only lists general articles, but indexes articles according to which machine they can be applied. Yes, there's even listings for the Heath computers and related articles.

The price on this index is \$14.95.

SOFTWARE FROM CDB

From California Data Base/ 4554 Cristy Way/ Castro Valley, CA 94546 comes some new software for your system. All programs require H89 or H8/H19 with single disk drive, HDOS and MBASIC. **TEST YOUR LOGIC** is a series of quizzes designed to test your logical ability. The system saves your scores on each test, allowing you to improve your score the next time thru. It costs \$19.95.

APTITUDE TEST consists of a set of tests which are used to determine your most likely aptitude. Suggested professions are included. Sells for \$39.

INSIGHT is a series of tests to help you develop and improve awareness of yourself. It will help you find out if you are realistic, uninhibited, adaptable, easily frustrated, etc. 2 diskette package for \$79.

THE MYSTERY WRITERS ASSISTANT is a totally new concept in computer software, designed to build a plot outline for Mystery Stories using a data-base of approximately 2000 entries. The system randomly selects an item from each of 11 categories. The output provides info such as: Where the crime was committed, Who the victim was, the Major clue, the Motive, Method of investigation, etc. Using the presented outline, the user can develop a complete mystery story. Cost on this is \$39.95

Last is **THE PERSONAL BUDGET SYSTEM** for \$29.95. It is an iterative system of budgeting, providing cycles as weekly, bi-weekly, semimonthly, monthly and annually. The system allows you to enter some income as weekly and some as monthly etc. Expenses can be handled in the same way and reports can be prepared in any desired cycle. Files can be saved for monthly reconciliation if desired.

SOFTWARE FROM ROBBINS

From **Steven Robbins/** 4610 Spotted Oak Woods/ San

Antonio, TX 78249 comes two HDOS software products. These will be reviewed at a latter date.

EDIT19 is an editor designed for the H89 or H8/H19. It is easy for the beginner to use. Documentation available in the 55 page manual discusses commands and tutorials for beginner and advanced user. The two modes, **SCREEN** and **COMMAND** control the editor. In **SCREEN**, the cursor can be moved anywhere on the screen so changes can be made directly to the file. In **COMMAND**, over 100 commands with easily remembered mnemonics are available. All standard editing features are provided, and some special ones, including: ability to handle lines of arbitrary length, handling of non-printing characters, listing directories and resetting disks, reading and saving parts of files, moving parts of files, searching only specified columns, setting the 8 special function keys to your desired commands, text formatting and right justification, on line documentation for all commands without leaving the editor, and a very extensive macro facility allowing storing sequences of commands on disk. There is too much to list here. Send a SASE for more info. The manual with a demo disk can be had for \$10, which can be applied to the purchase price of **EDIT19** within 90 days. **EDIT19** supplied on a 5" hard sector disk sells for \$75, complete with postage.

TIME is another product for use with HDOS 2.0. It allows the directory to store the time of file creation as well as date. Utilities to **SET** and **DISPLAY** the time are included. No hardware is included, but since source code is included it can be interfaced to a RTC. Note that it can be used with it's own device driver without any hardware clock present, although the accuracy will not be obtained. Utilities include a **PROLOGUE.SYS** for initializing the time on boot up and programs to display, set, start, stop, disable, increment, and adjust the rate of the clock. Software for doing this for the **WISE** RTC is included on disc. This also on 5" is \$20. For more details send SASE.

MULTI PURPOSE DEVICE DRIVER

From **Jim Teixeira/** 35 Shadow Oak Dr/ Sudbury, MA 01176 comes a truly universal device driver for printers, **UD.DVD**. It's quite unique as it will operate with virtually any printer including the MX-80 and H14, through its settable handshaking and format options and it also allows for up to 8 different independently configured unit numbers (like LP1:, LP2:, etc). For example, LP0: could be an H14 at 4800 baud and 80 characters/line. LP1: could be the same, but 96 characters per line. LP2: could be a Diablo on port 320 at 1200 baud, and so forth, with just one copy of the driver on disk or in RAM!!!

Some features include addressing any port between 300 and 377. Any standard baud rate between 110 and 19200. Variable left margin settings. Form and Page size between 0 and 255. Selectable lower to upper case mapping. Multiple copy printing. Selectable page titling and numbering. Selection of an optional ASCII string to initialize printer. A total of 17 options are settable for each of the eight units. The **HELP** command lists the present settings of each options for each unit. For those using **SYSMOD2**, a patch is included on disk to expand the **PR** command to add PR0 to PR7. The price on this is only \$29.95.

As mentioned in past issues of **H8SCOOP**, other products available from Jim include his **SYSMOD2** for \$17.95, **HFORTH** for \$29.95, **DIAS** (which includes a copy of Tiny Pascal) for \$25, Spanish, French, German and Italian Hangman for \$29.95, **BASEDIT** for \$15, and **EDGE** for \$25. For more information on any of these, or to order, contact Jim. He also says we can look for **CALC89**, which is a spread sheet that will rival **SUPERCALC**, which operates under HDOS, and **FLEX89**, which is a word processor for HDOS that is powerful and easy to use. These should be available early this year.

TALKING ADVENTURE GAME

More from MI-8/ 822 E. County Rd 30/ Ft. Collins, CO (303) 669-4116. **QUEST** is a talking adventure game, and sells for \$19.50. It comes on a 5" disk with documentation. System requirements are H8/H89 with 40K RAM, 1 disk drive, V-8 Voice synthesizer, and V-8 device driver (VB.DVD).

QUEST is a search for a pirate's hidden treasure which takes you through caves, caverns, and mazes. After locating the pirate's chest of gold, you will soon discover that getting it to safe keeping may not be so easy. About the same time you think you've finally made it, the old "Pirate" himself may suddenly appear to reclaim your prize, leaving you empty-handed to start your search again. The longer you play, the more it talks.

More software from MI-8 is on the way. Soon to be released is a direct text to speech device driver with more dictionary, making the V-8 easiest to program and most versatile Voice Synthesizer available. For more info on the V-8, see September and October 1981 H8SCOOP's.

FROM SOFTWARE DIMENSIONS

From **SOFTWARE DIMENSIONS/** 1061 Boot Rd/ Downingtown, PA 19335 (215) 269-7870 comes two new software packages. **MORSE CODE TUTOR** for \$19.95 is a program to help you learn or review Morse code. Whether you're a proficient Ham radio operator wanting to brush up on your skills, or a beginner trying to learn the code, Morse Code Tutor will fit your needs. With four selectable speeds of 5 to 13 words per minute it can: Translate a message you type in, into code using your computer's speaker; Provide random characters of code that you try to translate; Introduce you to seven different lessons where you are given five characters whose code sounds similar. This is done so you can distinguish these characters before moving onto others that also sound alike.

FIRST STRIKE is an exciting action graphics game that puts you in control of three anti ballistic missile bases trying to fend off nuclear and cruise missiles and thus save the east coast from nuclear annihilation. This program sells for \$14.95.

SOFTWARE FROM GENERIC

From **David Powers of GENERIC SOFTWARE/** POB 1154/ Troy, MI 48099 comes **CARD-IT**. It is a complete credit card management program that can save you time and money. **CARD-IT** maintains a list of the phone numbers you need in case your credit cards are ever lost or stolen. It also keeps track of the finance charges on your credit cards so you know how much interest to deduct on your income tax. **CARD-IT** indicates to you what your current credit card debt is, and how close you are to your credit limit. The following data is maintained: Credit card name, number, expiration date, credit limit; Phone number to call in case of card loss or theft; balance for each card; YTD Finance charge for each card; and total Finance charge (YTD) for all cards.

CARD-IT is completely menu driven, and has many more features. It requires H89 or H8/H19 with single drive and 48K HDOS or 56K CP/M to support up to 50 credit cards. It also requires the MBASIC interpreter. Price is \$24.95, specify HDOS or CP/M. Supplied on 5" disk.

Another software package, **FORM-IT** is a data entry forms package for the H19 which can enhance any data entry application. **FORM-IT** consists of a unique screen form pre-processor called **FORM-GEN**, and a set of MBASIC subs. It will take any user defined form up to 24 lines and create a screen parameter file for your application (sounds similar to **BASGEN**'s screen generator capabilities--ed). **FORM-IT** is useful for adding,

modifying, and displaying records in a database application. It requires H89 or H8/H19 with single drive and minimum of 32K. Price is \$19.95 and is supplied on 5" specify HDOS or CP/M. Contact **GENERIC SOFTWARE** for more info or to order.

MX-80 DRIVER

RAYMOND THOMPSON/ 12260 Welcome Dr/ San Antonio, TX 78233 (512) 655-3215 has available a **DRIVER** for the **MX-80**. This driver will support all Epson functions including graphics, and 4 configurations. This allows the user to set one configuration for 8 lpi, and one for 6 lpi. Either can then be accessed by EPX:. It is compatible with 1.6 and 2.0 HDOS. Cost for this is \$10 if you supply disk, or \$15 if Ray supplies disk.

DATA BASE PROGRAM

From **SUNFLOWER SOFTWARE/** 6304 W 76th St/ Prairie Village, KS 66204 (913) 648-1333 comes a new software product called **CONTROL**. It is a new database language compiler designed for data entry and retrieval. Coding is accomplished with easy to understand commands and directives. **CONTROL** allows you to create your own business application programs in a short time. It includes Full screen input and editing, Random access to data sets, full business arithmetic, conditional testing, and more. It does not say the operating system, but I assume HDOS. Cost is \$99.95, which includes binder with 43 pages documentation.

Besides this, they also have **DUALPORT**, **EASY**, **SAP**, **MAILMATE**, **DISKSORT**, **BILLING**, and **NAMELIST**. Many of the Heath stores carry their products. To receive more information or to order, contact Rick Kerbel of Sunflower.

PROOF READING PROGRAM--THE WORD

From **OASIS SYSTEMS/** 2765 Reynard Way/ San Diego, CA 92103/ (714) 291-9489 comes **THE WORD**. **THE WORD** is a spelling correction program for CP/M systems. It will proof read your text and interfaces with most CP/M word processors, including Spellbinder. It was reviewed in November **BYTE**, page 434 along with 4 others, and got the highest ratings and yet has the lowest price of \$75. It requires CP/M operating system with 32K and is distributed on 8" single density disks. As with any proof reading program, because of the large dictionary, it will not work on 100K drives. The main dictionary alone, is 132K. It would possibly work on the 400K Tandons if downloaded to that format.

This is the first issue of this newsletter that is being proof read by computer, thus from here on, my spelling should be improved. I am using **THE WORD**, and am very much impressed!

LAND MOBILE RADIO PROGRAMS

FRONT RANGE COMMUNICATIONS, INC./ Box 12010/ Bolder CO 80303 (303) 443-6100 is introducing a line of computer products for the Land Mobile and related industries.

EMI for \$49.95 is an electromagnetic interference program used to predict radio frequency interference possibilities at radio communications sites. It will find and define intermodulation products to the ninth order for two and three transmitter combinations, and harmonic combinations to the sixth order. It accepts variable bandwidths and up to 100 transmitters and 100 receivers. Requires CP/M, disk drive, and Z80 CPU. Available on 5 and 8" disks.

TOTALREC is an accounts receivable program group that will print your invoices, update your receivable file, age them, issue credit memos, keep track of who paid, how, when, and more. Requires CP/M, 200K disk storage, 48K memory, Z80, and printer. 8" or Winchester storage is

recommended. Price is \$79.95 available on 5" or 8" disks.

HORIZON is an interactive program that gives the communications engineer path loss info on VHF and UHF radio paths. Enter the geographic coordinates, frequency, antenna gain, feed line loss, and "K" factor. **HORIZON** tells you the distance, direction, path loss, antenna height required. Selling for \$49.95, it requires CP/M, Z80, 48K, and 100K disk storage. Available on 5" or 8" disks.

TAPE BACKUP UTILITY

Available from **KANDUEAZY COMPUTER SOFTWARE SERVICES/** 6218 Blossom Ln/ Alexandria, VA 22310 (703) 922-9450 is a CP/M file to Tape utility called **TAPEIO**. It is a multifunction utility allowing the CP/M user to interface an H-8 (with H8-5 serial/ cassette card) with a cassette tape. Various commands available are:
BACKUP--backup CP/M disk file to tape.
RESTORE--restore disk file from tape file
TEXT--read a text file created by TED-8, uncompress it, and write it to disk.
VERIFY--check tape file for checksum errors or missing blocks.

This program eliminates the need for backup disks. All those TED-8 tape assembly language programs or text files you wanted to use but did not because of the time required to re-key in everything are now a simple command away, with **TAPEIO** doing all the work in a few minutes. Complete error checking when reading a tape file.

For a copy of the doc, send a legal size SASE to Kandueazy. To receive your copy of the **TAPEIO** utility, send \$25 (or \$22 plus disk). Heavily documented source code is also available for \$35.

Also available are high quality leatherette dust covers for H8, H9, H17 and H19 (which includes H88 and H89). They are \$12 each plus \$2 postage and handling. Send for a sample of the material used by sending SASE.

James Kandul of Kandueazy also has a Tiny Pascal compiler written in Heath's tape extended BH Basic. If you have an interest, let him know and he will write some documentation for it and make it available.

MULTITUDE OF SOFTWARE FROM VIKING

If you want software and utilities to do just about anything, send for the product list by **VIKING SOFTWARE** (Grant Gustafson)/ 2243 S. Belaire Dr/ SLC, UT 84109 (801) 484-9573 (ask for Grant). You'll receive over 10 pages of product descriptions, including the specialized Device Drivers mentioned in Past H8SCOOP's.

They have come out with some new products. They are too numerous to discuss here, I'll just mention them, and if interested send for the product list. **SHELL-METZNER** Sort Device Driver for MBASIC is unique as the **SORT** routine is a machine language program loaded and called as a device driver. Since it is a device driver, it can easily be called from MBASIC. Since it's .ABS, its very fast.

LPIP.ABS is a librarians PIP program, a file based copier for HDOS. It provides the services under HDOS currently provided on DEC mainframes for filelist transfers under the **DECSYSTEM COPY** command. An example of its use would be to put selected files from an 8" disk to a 5", without ever typing a file name. Many other features.

CRASH.ABS for \$19.95 will work with the 100K or 400K drives, CP/M or HDOS, mountable or otherwise. This is the only disk save utility that salvages the remaining good sectors from an unmountable disk with no track 0, no directory, and no GRT table, with some degree of automation.

COMP.ABS for \$9.95 is a compare utility and patch

documentor. It accepts file names on the system, reads both files to memory and prints out any differences. Along the same lines is **PIRATE.ABS** for \$19.95, a 100K whole disk compare utility for use with the 100K drives. It can handle CP/M or HDOS. Any differences in the two disks can be directed to TT, LP, or any HDOS file. Can be used to locate passwords, ID numbers, and system data locations.

XCOPY.ABS for \$9.95 is a compact 4 sector 50 second 100K disk copier for use with HDOS disks. **XCOPYCPM.ABS** is the same thing, running under HDOS, but used for copying CP/M disks. Supplied with **XCOPY** free.

DISKTAPE.ABS is a 2 way cassette tape archival utility for \$24.95. It is compatible with **DSKCAS** offered by HUG. It runs on the H89 with cassette card, or H8 with H8-5 serial/ cassette card. It will fetch from tape, or write to tape, all in one stand-alone program. To go with this is **DSKBATCH.ABS**, an auto dump from disk to cassette. It reads the directory, prints the contents, and lets you select the files to be dumped to tape. A whole disk can be dumped by a few keystrokes, without operator intervention during dump. Files marked with the "S" flag are not dumped. A 90 minute tape should hold about 10 diskettes or more. **TAPBATCH.ABS** is auto dump from cassette tape to disk, or the reverse of **DSKBATCH**.

DIAL.ABS modem for H89 or H8 with H8-4 serial board sells for \$29.95. The H19 connected to a mainframe is a superb workstation. Adding a micro opens the possibility of storing the session in memory or disk. This modem program allows an instant toggle between micro, and a standard video terminal wired into the mainframe. This software interface is in every sense a replacement for a \$5000 intelligent downloading terminal. Regular and auto answer modems are supported. Perfect for **SOURCE**, **MICRONET**, **DIALOG** and other systems. It also supports the H9.

SND.ABS for \$19.95 is a high speed Heath to Heath file uploader. It performs the Heath to Heath file transfers with protocol and error handling made famous for HDOS by the well-known Softstuff program **CPS**. **GET.ABS** which is included, is a high speed Heath to Heath file downloader.

HDOS 2.0 OVERHAUL SOURCE DISK for \$9.95 requires the latest version of Jim Teixeira's **SYSMOD** to be installed. It consists of a collection of patch files which can be installed using **DUMP** and/or **MBASIC**.

Coming early this year is a universal H89 joystick card for all video games, and a joystick driver. Games like **Invaders**, **Munchkins**, etc, will interface to the joysticks with under 3 minutes of setup time. With all products, the media charge is extra, and so is postage. This lets you select media and method of delivery. Contact **VIKING** for more information on any of this, or to order.

CP/M UTILITIES FROM ELEKTROKONSULT

Terje Bolstad of Elektrokonsult/ POB 846, N-3001 Drammen/ NORWAY has two excellent utilities available to work with CP/M. They are probably the only drive independent disk utilities of the kind for CP/M. In other words, they work with any drives and disks by automatically setting themselves for the media being used.

DDUMP is a sector oriented disk dump utility, which makes it possible to examine and modify any byte(s) on any sector, addressed by track and sector, or allocation block number. Sector contents are displayed (and/or printed) and may be patched in HEX and ASCII formats. **DDUMP** allows you to dump and patch data not otherwise accessible by CP/M. It is a very complete dump. I have tried it on my 8" DS DD Magnolia system with no problems. It is very versatile and has a Help command, in case you

forget what the commands are. Comes with complete easy to follow documentation, which includes sections on CP/M logical and physical sectors, disk characteristics, and how a disk is used by CP/M. This is useful for those who want to see where CP/M puts everything, and what the disk structure is. It even contains hints on how to recover deleted files, provided the directory is intact. DDUMP is made to dump sectors, it cannot dump a file name, use DUMP for something like this.

DTEST is a fast test utility which tests a disk by writing and reading a test pattern on every sector of an initialized disk. Disk errors are listed to console and can be listed to printer also. They are listed with the track, sector, and allocation block number. All bad sectors are automatically locked out from later use by CP/M. This is a good method to test your CP/M disks, and recover those that had bad sectors. DTEST also works on any format drives and disks. I have used it with my 8" with no problems. I had a bad disk which I ran through DTEST. It did show up bad. I re-initialized it, ran DTEST again, and all was well. Evidentially, the init E5's were destroyed on some sectors, or never got there.

There are various tests DTEST can perform, from 1 to 99 times. You can read only, which is fast, or write and read, which is slower, but more complete. It is also said to be able to test disk drives. By running tests on different disks for a few times, the same errors in the same places, would indicate defective disk drive. Like DDUMP, it is easy to use, and comes with complete easy to follow and understand documentation.

Both programs are highly recommended to CP/M users. Price is \$29.95 each, which includes disk and documentation. Source code NOT included. Disk format 5" Heath CP/M. Check or VISA accepted, include \$4 per order for postage and handling.

MORSE CODE PROGRAM

MLM MORSE is an easy to use Morse code transceiver program for H8 and H89 owners interested in digital communications. It converts International Morse code from a receiver into an alphanumeric video display and changes characters typed at a terminal into the form needed to activate a transmitter or a code practice oscillator. A split screen at line 16 separates the received and transmitted portions. Program parameters are controlled by the console special keys. Shifting between receive and transmit modes is accomplished in an automatic fashion.

The problem in any Morse program is the hardware interface. A variety of options are available. You can build your own from the supplied information. If you have an RTTY modem you can use it to receive CW as well. MLM can supply you with a complete hardware and software package which includes the program, an MFJ-1200 computer interface, and power supply for \$99.95. Deduct \$10 if you will furnish your own power supply. Program alone is \$29.95. To order, or for more info, contact William Hall/ MLM Associates/ 5621 Maple Heights Ct/ Pittsburgh, PA 15232.

PERSONAL FILE MANAGEMENT

Personal File Management (PFM) is a small scall database file handler for use by the hobbyist. It will assist you in keeping your files in an orderly, sequential format, while still maintaining easy data access. PFM will create a database file, update the file, delete records, search for records, add records, sort on any field, list the file in any format, and sum or average any numeric field. Suggested uses range from an inventory data base to a magazine article index, or from a customer history file to a mailing list. You are limited only by your imagination.

Requirements are H8 (with H19) or H89 with 48K memory and 2 disk drives are recommended. MBasic 4.82 or 4.70 is

required, and the version must be specified when ordering. The price is \$50 and it is available from HAT SOFTWARE/ POB 27402/ Honolulu, HI 96827.

HOUSEMASTER

ARTRA INC/ POB 653/ Arlington, VA 22216/ (703) 527-0455 is announcing HOUSEMASTER--a new PC card for the H89. The standard HOUSEMASTER board provides the H89 with four separate peripherals on a single card, including voice recognition, stereo sound synthesis, a real time clock/ calendar, and a BSR-X-10 home control interface. Available options include battery backup for the clock, two types of voice synthesis, and two RS-232 serial ports all on the same board. This card plugs into the right side of the H89.

Included is a disk of 17 programs, along with a 90 page manual. The programs demonstrate all of the features of the board. One of the programs is the most sophisticated control program available for any personal computer. By combining all four features of the board, the program provides complete control of household devices by voice, date and time, or keyboard. User developed subroutines can be easily added and controlled in the same way.

As an example, the program could be tailored to wake you up at 8AM with chimes and then start voice control. You could then tell it (by voice) to turn on the lights in your bedroom, start the coffee pot, and turn on the stereo. With the voice synthesis option, the computer can even answer your questions with its own voice. The price of the Housemaster board is \$299 kit or \$399 assembled, calibrated and tested. Battery backup option is \$25 kit or \$35 assembled. Add \$3 shipping per order, and sales tax where applicable. COD's are accepted. For a brochure with more details, or to order, contact ARTRA INC.

BIBLIOGRAPHY UPDATE

I'm glad someone gets Interface Age and keeps up with goodies, as I no longer receive it. Clem Peper sent me a copy on November 1981 Interface Age, page 112 where an article SHOW A CP/M PROGRAM TO DISPLAY ASCII FILES was presented. Clem said he typed it in just as it was and it ran perfectly. It is used in place of the CP/M TYPE command to display ASCII text on the screen in 22 line segments. To get the next 22 lines, simply tap the space bar. Tap any other key, and you are back in CP/M command mode.

REQUESTS

Chris Simmons/ 2802 N Greenwood St/ La Grande, OR 97850 has the following request. If you can help him, please contact him. He wants to use the Heath UCSD Pascal with his Heath H8-2 Parallel board. Presently, the only interface board supported by UCSD is the H8-4 serial board. Chris has a parallel printer connected to his H8-2 and would like to drive that via the Pascal.

Feedback from the Request by George Duke about the Detroit Metro HUG as printed in issue #18 (Sept 81) H8SCOOP: It appears Detroit Metro HUG has had organization problems the newsletter has slipped. They have offered to refund the money George paid for his subscription he never received. They do not plan to make subscriptions for non members available in the future, because of all the hassle it caused.

A question that is asked me again and again, (I ask myself also) and the answer has not thus far been provided (at least not to my satisfaction). Under HDOS you can access all kinds of printers and devices very easily through the device drivers. Under CP/M, you have your list device (usually a printer) and that's it! Is there any (easy) way you can access say a line printer and an alternate terminal under CP/M and programs running under CP/M? I've been told that the ONLY way this is possible is to modify the BIOS (that means assembly language and time, and

knowing what you're doing--that ain't easy folks!). Is that true? If anyone has first hand knowledge on this, please write in and let us all know.

Frank Pfiefer/ Cicero School/ 23rd Street & 49th Av/ Cicero, IL 60650 has a BASE 2 Model 800B Printer. He wants to know if anyone has a good device driver for this printer using the H8-2 parallel card. He eventually wants to get the H8-4, but for now is interested in the parallel card. Anyone do it?

Ken Jean/ 1300 Lafayette East, Detroit, MI 48207 wants to know if anyone has ever heard (or know) anything about CFR Associates, Inc. of Haverhill, MA? He continually sees their adds in Kilobaud for used Selectrics and Daisy wheel printers with reasonable prices. Is it too good to be true?

D.C. Shoemaker/ 2000 A Foxridge/ Blacksburg, VA 24060 would like to know if anybody knows how to turn off and on the H8 front panel from MBASIC. In BH Basic CNTRL 2,0, CNTRL 2,1, and 2,2. Using POKes there should be a way in MBASIC. Anyone have an idea? I think UO.NFR, which is at 040.010 can be set to cause front panel display, and UP.DDU 000.002 is used to disable the display update, bit 2 set is disable, clear is enable. In fact, there are a number of user option bits (UO.) that can be set in .MFLAG (000.xxx) for various H8 front panel. UO.HLT is 000.200 for disabling halt processing. 000.100, UO.NFR, is for no refresh of front panel. UO.DDD, Disable display update is 000.002, and Allow Private Interrupt Processing, UO.CLK, is 001. .MFLAG address is 040.010. There are several interesting ones for HORN, and Front Panel LED display. Look at your assembly listing of XCON or PAM for locations and bits to set--ed.

OFFERS

I make all monthly software available on an as-is basis, in other words, no money back policy if not satisfied, mainly because it's too easy to copy discs. You take your chances. At my prices, you really can't go wrong. I try to get software out as fast as possible so you can get the newest stuff fast, and as thus, don't get a chance to carefully screen and test everything. To do so, would delay all software at least 3 to 4 months. I do the best I can and sell without assuming responsibility on the accuracy or runability of a program or disc. I don't think I've let anyone down in the past, and will continue to do my best not to.

The following are past offers made by H8SCOOP, which are and will continue to be made available to those interested readers. Note I have dropped a few items from the list, and as time goes on, more will be dropped so my inventory doesn't go crazy around here.

Tandon Drive W/ HDOS Software	\$460
Tandon Drive only	\$425
2ND. Drive	\$415
HDOS 400K DRIVER	\$ 40
HDOS MAGNOLIA DD DRIVER	\$ 40
CP/M BIOS 80 DRIVER	\$ 30
H89 DD Magnolia board W/CP/M	\$505
SPELLBINDER Office Management Pkg.	\$295

PROJECT ARTICLES FOR PARALLEL PORT--WRITE FOR CATALOG

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FEBRUARY issue deadline--January 15

Subscription Rates--\$20/Year (\$27 overseas Airmail)--current issues

Back issues available start with Volume 2, issue # 13. Volume 2 back issues available at \$1.25 each to current issue.

PHONE HOURS: 3AM to 9PM as I am available, except sunset Friday to sunset Saturday. NO COLLECT CALLS.

9/80 File Dumps & Utilities.	\$10
11/80 Pictures, Lables, Word Processor.	\$15
12/80 Pictures & Utilities.	\$10
1/81 H19 Graphics program disc.	\$10
2/81 Plant, Swarms, Starwars, Utilities.	\$10
3/81 Potpourri Set(2).	\$15
3/81 Adventure Set(2) Pirates, Lost Dutem	\$15
4/81 THINGS--Graphics, Biorythm, Utility	\$10
5/81 HDOS 2.0 PATCHES	\$10
7/81 MISC I--OREGON TRAIL, ATOM20	\$10
7/81 ODDS & ENDS--Sort, Calculator, Finance	\$10
8/81 HVC; Bigfont (excellent), cake, chair	\$10
10/81 MIXED BAG--400K DUP & DUMP & MISC	\$10
11/81 MULTIKEY SORT	\$6

Because of lack of time, and lack of contributed software, there will be no offer this month--sorry.

CLASSIFIEDS

FOR SALE--A/J 841 Selectric Terminal. Serial interface, 1K buffer, selectable BAUD rate. Currently running with H8. \$800. Contact Richard Hartman/ (213) 564-3075 evenings, or 564-2591 days.

FOR SALE--Disk drive power supply assemblies. These are the same units I have been selling with the cabinet for the Tandon Drives. I have about 4 left, 5V and 12V at 1A each. Ready to go, with metal back cabinet bracket. I'll sell them for my cost to clear them out \$45. Contact H-SCOOP.

FOR SALE--WH8-16 16K RAM for H8 for \$175 each. Contact Howard Schuster/ 140 West St/ Room-2910/ NY, NY 10007, phone (212) 394-1190 from 9-5.

WANTED--Used Z89 computers for under \$1500. Contact Allen Bernson/ POB 68/ Boston, MA 02167.

FOR SALE--Three Heath WH16 Static RAM boards for \$150 each. Bill Bechtold/ 7429 Frederick St/ Omaha, NE 68124. Phone (402) 397-2461.

FOR SALE--Four Heath 8K RAM boards in excellent condition, for \$65 each. Contact Van Christopher/ 9602 Pebble Beach Dr/ Santee, CA 92071 (714) 448-9223 evenings.

FOR SALE--One H-8 digital computer setup with 16K memory. \$600. Contact John Kofroth/ 25 Concord Rd/ Darby, PA 19023. Phone (215) 237-6314 days (home), or 496-4000 Ext 5087 night (work).

FOR SALE--Trionyx 64K memory board fully socketed with 48K RAM, \$300. H8-4 four port serial board \$140. H8-2 3 port parallel board \$100. H8-5 serial cassette board \$80. H-17 (steps at 6ms) for \$425. All above with gold connectors. Contact Bill Serp/ 4027 Harvey Av/ Western Springs, IL 60558/ (312) 246-7856.

FOR SALE--WH8-16 16K RAM, \$175. Three H8 8K RAM's, \$60 each. Jack DelPorte/ 2226 NE 19th Ct/ Jensen Beach, FL 33457. (305) 334-0140 home; 287-5300 X 211 office.